Supplementary material for
A spectral theory for turbulent burning velocity across scales

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This document accompanies the paper. Sections, figures, tables and equations numbered with the prefix S belong to this document; all other references are to the paper. The figures are produced by the notebook `make_figures.ipynb` from the same data as the figures of the paper.

S1 Distribution of the wrinkling exponent

The paper connects its burning-velocity relation with the power-law wrinkling factor of Charlette *et al.* (2002*a*) for an exponent $\beta_w = 1/2$. Charlette *et al.* (2002*b*) determined β_w locally with a dynamic procedure, and figure S1 shows the probability density function (PDF) of the exponents they obtained, digitised from figure 5 of their paper. The digitised curve, which ends at $\beta_w = 1$, was normalised to unit area and its moments were computed by trapezoidal integration. The mean is 0.587 and the standard deviation 0.104 (variance 0.0109); the median is 0.57 and the interquartile range 0.51–0.65. About 22 % of the distribution lies below $\beta_w = 0.5$, and the digitised PDF falls to zero at $\beta_w = 0.4$, above the fractal value $\beta_w = 1/3$ that would recover Damköhler's large-u' limit $s_T \propto u'$. The distribution therefore supports an exponent close to $1/2$, centred slightly above it.

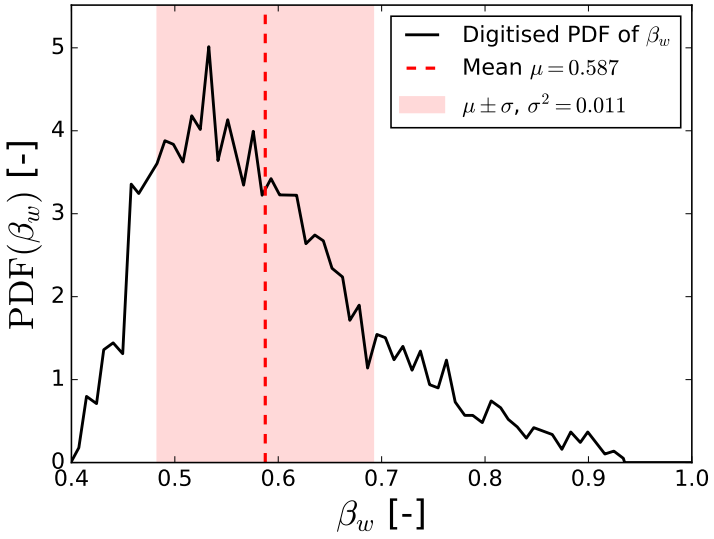


Figure S1: PDF of the local wrinkling exponent β_w obtained with the dynamic procedure of Charlette *et al.* (2002*b*) (digitised). Dashed line, mean $\mu = 0.587$; shading, $\mu \pm \sigma$, where $\sigma^2 = 0.011$ is the variance.

S2 Selection and treatment of the experimental data

Measured turbulent burning velocities depend on the flame configuration and on how s_T is defined (Driscoll, 2008), so the datasets of table 1 of the paper were chosen to keep both as uniform as possible.

S2.1 Flame configuration

Driscoll (2008) groups premixed turbulent flames into envelope flames (on two-dimensional slot and axisymmetric Bunsen burners), oblique flames, unattached flat flames (on low-swirl, diffuser or counterflow burners) and spherical flames (in fan-stirred vessels). All of the datasets except Aachen are envelope flames: the Michigan (2005) flames are on a slot burner, Damköhler's on tubes and the others on round Bunsen burners. The Aachen flames, on a low-swirl burner, are unattached flames in this classification.

S2.2 Turbulent burning velocity

In an envelope flame, all of the reactants supplied by the burner pass through the flame, so mass conservation gives a mean, or global, consumption speed,

$$s_T = \frac{\dot{m}_R}{\rho_R \bar{A}}, \tag{S1}$$

where $\dot{m}_R$ is the mass flow rate of reactants, ρ_R their density and $\bar{A}$ the area of a mean flame surface (Filatyev *et al.*, 2005; Driscoll, 2008). Where the source reports it, $\bar{A}$ is taken on the isosurface $\overline{\Lambda} = 0.5$ of the mean progress variable (table 1 of the paper); for Ottawa it is $\overline{\Lambda} = 0.05$, and for Michigan (2005) it is the time-averaged flame position. Differences in equivalence ratio, within and between datasets, are accounted for by normalising s_T by the laminar burning velocity s_L of each mixture.

Spherically expanding flames, such as those studied at Leeds (Bradley *et al.*, 2003; Lawes *et al.*, 2012) and Princeton (Chaudhuri *et al.*, 2012; Wu *et al.*, 2015), are characterised instead by a propagation or displacement speed derived from the growth rate $\mathrm{d}R/\mathrm{d}t$ of a mean flame radius R, often multiplied by the ratio of burned to unburned gas density. This speed increases with R, because a growing flame kernel is wrinkled by an increasing part of the turbulence spectrum (Abdel-Gayed *et al.*, 1987; Bradley *et al.*, 2003; Lawes *et al.*, 2012), and Chaudhuri *et al.* (2012) correlated it with a Reynolds number based on the mean flame radius rather than on ℓ_0. It measures a different quantity from (S1), and such data are not included.

S2.3 Approaching turbulence

The approaching turbulence differs between the configurations. In an envelope flame, it is generated in the burner tube or nozzle, typically by a perforated or slotted plate or a grid, and in Damköhler's experiments it is the turbulence of the tube flow itself, so its spectrum is that of a turbulent pipe or nozzle flow. Pope (2000, figure 6.14) derived the model spectrum used in the paper with one-dimensional spectra measured in a range of flows, among them turbulent pipe flow and round jets. In fan-stirred vessels, the turbulence is produced by fans and is close to isotropic near the centre; its spectrum has been characterised for the Leeds vessel by Abdel-Gayed *et al.* (1987), who measured its power spectral density, and in Walker (2019).

S2.4 Mixtures and operating conditions

To exclude differential diffusion, all of the datasets except Damköhler's are methane–air flames, for which the Lewis number $Le \approx 1$ (e.g. Kobayashi *et al.*, 1996), at 0.1 MPa and about 300 K; for this reason, the high-pressure measurements of Kobayashi *et al.* (1996) and the propane–air flames of Yuen & Gülder (2013) are not used. Damköhler's flames burned a propane–butane mixture with oxygen.

S2.5 Laminar flame thickness

The sources define the laminar flame thickness differently. Michigan (2017) and Toronto (2013) give thermal thicknesses from detailed-chemistry calculations of the laminar flame (0.36 mm, and 0.45–1.0 mm for the flames used here); Michigan (2005) uses $\delta_L = 7.4\, D_{\mathrm{th}}/s_L = 0.35$ mm, where D_{th} is the thermal diffusivity of the reactants; Aachen uses $D/s_L = 0.35$ mm with the diffusivity D evaluated at

1500 K; and the thicknesses of Tohoku and Toronto (2014), 0.066 mm and 0.055–0.11 mm, correspond to D_{th}/s_L. Ottawa gives 0.033–0.065 mm. The paper takes ℓ_0/δ_L as given by each source, or as computed from the given ℓ_0 and δ_L. With unity Schmidt number, a consistent choice would be $\delta_L = \nu/s_L$, so that $Re = u'\ell_0/\nu$. Reynolds numbers defined in this way, reported by the sources or computed from their tabulated ν and s_L, exceed those used in the paper by factors of 5.5, 6.6 and 8.9 for Michigan (2017), Toronto (2013) and Michigan (2005), and by 1.4 for Tohoku and Toronto (2014). The thickness enters the prediction only through $\mathcal{F}(Re)$, which varies slowly: halving or doubling δ_L changes the predicted s_T by about 10 % at the median and by at most 21 %. Re-evaluating Re as $u'\ell_0/\nu$ for these five datasets changes the fraction of points within a factor of 1.5 of the prediction from 71 % to 69 %, and the r.m.s. logarithmic deviation from 0.38 to 0.37, although individual datasets shift: the geometric-mean ratio of measurement to prediction changes from 1.06 to 0.96 for Michigan (2017), from 1.06 to 0.74 for Toronto (2013) and from 1.37 to 1.04 for Michigan (2005).

S2.6 Regime diagrams

Figure S2 places the datasets, with the same u'/s_L and ℓ_0/δ_L as figure 1(a) of the paper, on two variants of the regime diagram.

Skiba *et al.* (2018) measured preheat- and reaction-layer thicknesses in the Michigan flames and compared them with flame structures reported in earlier experiments and simulations. The Klimov–Williams criterion $Ka = 1$ did not separate thin flamelets from flames with broadened preheat zones, so they replaced it by a ratio of turbulent to molecular diffusivity, $D_T/D^* = u'\ell_0/(s_L\delta_L)$, which equals Re. Flames with $D_T/D^* \gtrsim 180$ had broadened preheat zones but thin reaction layers, and no broadened or broken reaction layers were observed up to $Ka = 533$. Their measured regime diagram (figure S2a) adds tentative boundaries at $Ka = 60$, which extends the broadened-preheat regime to small ℓ_0/δ_L, at $Ka = 1$ for $D_T/D^* > 180$, and at $Ka \approx 550$ for broadened reaction zones. Most of the present measurements fall in the thin-flamelet region; the broadened-preheat region contains the high-Ka Michigan (2017) and Toronto (2013) flames and part of the Aachen data, and no dataset reaches $Ka = 550$. Skiba *et al.* (2018) use the diffusive thickness of Peters (2000), with the diffusivity evaluated near the reaction layer, so the positions of datasets based on other definitions of δ_L (§ S2.5) are approximate.

The Bray number (Veynante *et al.*, 1997),

$$N_B = \frac{\tau s_L}{2\alpha u'}, \tag{S2}$$

compares the gas expansion across the flame, through the heat-release factor $\tau = \rho_u/\rho_b - 1$, with the turbulent velocity fluctuations; α is an efficiency of order unity. For $N_B > 1$, the expansion dominates and the turbulent scalar flux is counter-gradient, and for $N_B < 1$, it is gradient (Veynante *et al.*, 1997). A counter-gradient flux implies a negative turbulent diffusivity, and Nivarti & Cant (2017) found that this limits the validity of Damköhler's second hypothesis, on which the diffusive reading of the paper's relation rests. Figure S2(b) shows $N_B = 1$ on the regime diagram of figure 1(a) of the paper for $\tau = 6.67$, the value for the stoichiometric methane–air flames of Nivarti & Cant (2017). The efficiency is either $\alpha = 0.25$, which reproduced the transition in their simulations at $\ell_0/\delta_L = 2.75$, or increases logarithmically with ℓ_0/δ_L as $\alpha = 0.4\log_{10}(\ell_0/\delta_L) + 0.059$ (but not below 0.05), which passes close to $\alpha \approx 0.3$ at $\ell_0/\delta_L = 4$ (Kheirkhah & Gülder, 2015) and $\alpha = 0.5$ at $\ell_0/\delta_L = 11$ (Veynante *et al.*, 1997). With $\alpha = 0.25$, 17 of the 75 measurements with $u' > 0$ have $N_B < 1$, 13 of them from Michigan (2017), and the geometric-mean ratio of measured to predicted s_T is 1.12 on both sides of the line. With the logarithmic α, 29 have $N_B < 1$, and the ratios are 1.06 for $N_B < 1$ and 1.15 for $N_B > 1$.

S3 Scaling of the burning velocity with Reynolds and Karlovitz numbers

Figure S3 shows the measured s_T/s_L divided by Ka against Re for the eight datasets of table 1 of the paper, with the same values and symbols. The dashed lines are $100\,Re^{-n}$, labelled with their slope $-n$.

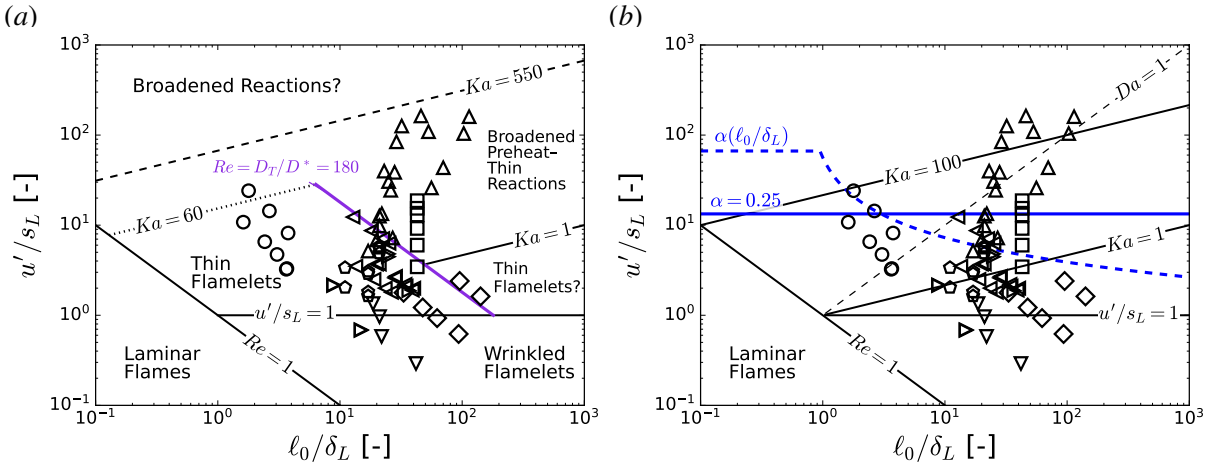


Figure S2: The datasets of table 1 of the paper (symbols as there) on (*a*) the measured regime diagram of Skiba *et al.* (2018), with their measured boundary $D_T/D^* = 180$ (purple) and tentative boundaries $Ka = 1$, 60 and 550; and (*b*) the regime diagram of figure 1(*a*) of the paper with the line $N_B = 1$ for $\tau = 6.67$ and $\alpha = 0.25$ (solid blue) or the logarithmic $\alpha(\ell_0/\delta_L)$ (dashed blue). Below the blue lines, $N_B > 1$ and the scalar flux is counter-gradient.

Since $Ka = (u'/s_L)^{3/2}(\ell_0/\delta_L)^{-1/2}$ and $Re = (u'/s_L)(\ell_0/\delta_L)$, a series of measurements at fixed ℓ_0/δ_L in which $s_T/s_L \propto (u'/s_L)^m$ follows

$$\frac{s_T/s_L}{Ka} \propto Re^{\,m-3/2} \tag{S3}$$

in this plane. A slope of $-1/2$ thus corresponds to Damköhler's linear scaling, $m = 1$, and a slope of -1 to $m = 1/2$, the strong-turbulence limit of the paper's relation $s_T/s_L = [1 + C\,\mathcal{F}(Re)\,u'/s_L]^{1/2}$ apart from the weak variation of $\mathcal{F}$ with Re. Datasets in which ℓ_0/δ_L varies together with u'/s_L do not follow (S3) exactly.

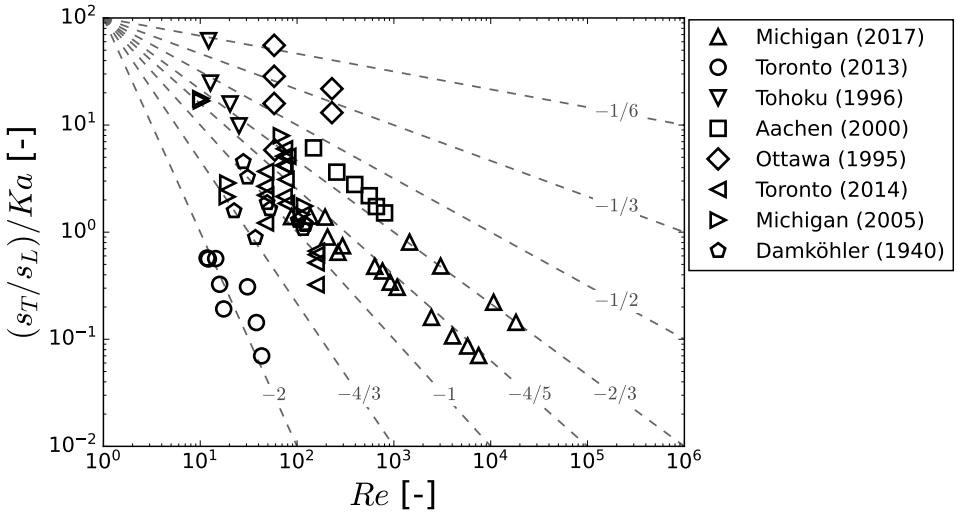


Figure S3: Measured s_T/s_L divided by Ka against Re for the datasets of table 1 of the paper (symbols as there). Dashed lines, $100\,Re^{-n}$, labelled with their slope $-n$.

S4 Fits to the spectral eddy viscosity and diffusivity

The functions $\nu_e^+(\kappa/\kappa_c)$ and $D_e^+(\kappa/\kappa_c)$ computed by Chollet (1984) were digitised for $0.01 \leqslant \kappa/\kappa_c \leqslant 1$ and fitted by nonlinear least squares with the form that Pope (2000) used for ν_e^+,

$$\nu_e^+,\ D_e^+ = C^{-3/2}\left[A + B\exp\left(-c\,\frac{\kappa_c}{\kappa}\right)\right], \qquad C = 1.5. \tag{S4}$$

Table S1 lists the coefficients and the values at the cusp, $\kappa = \kappa_c$, which are the values used in the paper; the turbulent Schmidt number is $Sc_T = \nu_e^+/D_e^+$. The fitted curves are shown in figure 2(b) of the paper. The regression of Pope (2000) for ν_e^+ gives a cusp value 26 % lower than the fit to the curve of Chollet (1984).

	A	B	c	Value at $\kappa = \kappa_c$
ν_e^+, Pope (2000)	0.441	15.2	3.03	0.640
ν_e^+, fit to Chollet (1984)	0.4927	162.7	4.996	0.868
D_e^+, fit to Chollet (1984)	0.8717	949.5	5.908	1.88
$Sc_T = \nu_e^+/D_e^+$				0.462

Table S1: Coefficients of (S4) and values at the cusp.

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